



Chapter Notes: Beyond Earth

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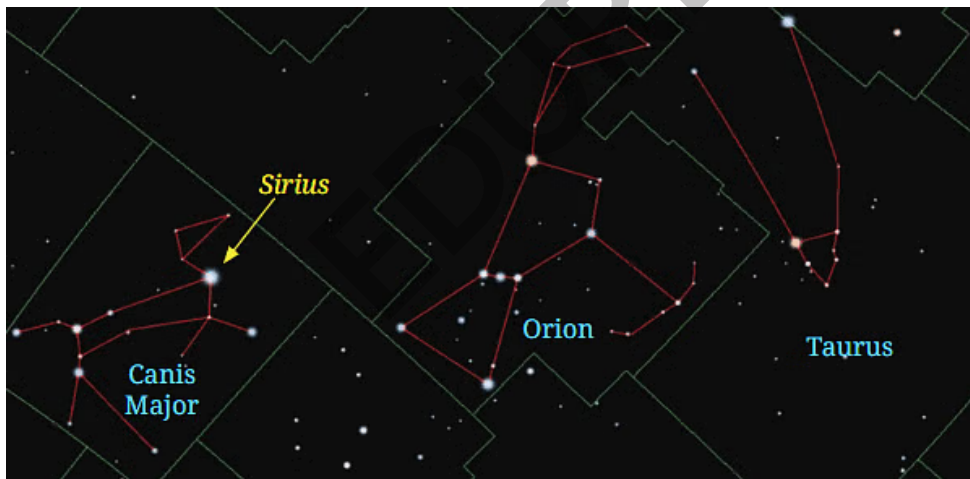
Introduction

Nubra, a picturesque region in Ladakh, is known for its majestic mountains, glaciers, and clear, cloudless skies. The nights in this region are especially captivating because the absence of air and light pollution allows thousands of stars to shine brightly. People have long been fascinated by the night sky and have asked questions about the size, distance and arrangements of the stars. In earlier times, certain stars helped guide caravans travelling through Nubra, showing that the night sky has both practical uses and the power to inspire wonder.



Stars and Constellations

Stars are large, hot bodies of gas that produce their own light and heat. At night they appear as points of light, some brighter and some fainter. When people look carefully at the sky, they often notice that certain stars appear to form familiar shapes. These imagined patterns are useful for remembering and locating groups of stars.



Stars produce their own light; their apparent brightness depends on both how much light they emit and how far away they are.

When a group of stars forms a familiar shape, we call that group a **constellation**. Different cultures gave these patterns names based on animals, objects or characters from stories. These star patterns made it easier to recognise parts of the sky and to pass on knowledge from one generation to another.

Stars and Navigation

Before the invention of modern instruments such as the **magnetic compass and GPS**, sailors and travellers used stars and constellations to find directions and navigate at night. Even today, star patterns can serve as a reliable backup method of navigation in emergencies.

MULTIPLE CHOICE QUESTION

Try yourself: What did our ancestors use stars for in ancient times?

- A To predict the weather
- B To navigate
- C To communicate with each other
- D To determine time of day

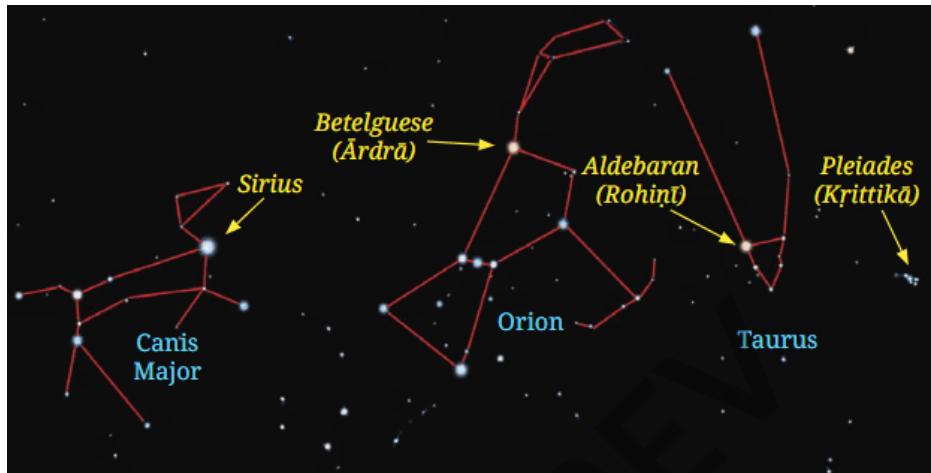
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Constellations

Groups of stars that form a recognised pattern are called **constellations**. To help see the pattern, people often draw imaginary lines between bright stars. For example, the constellation **Orion** is seen as a hunter, and its three bright stars in a row form **Orion's Belt**.

Orion's belt is formed by three bright stars almost in a straight line.

Different cultures imagine different stories: some see Orion as a hunter who is followed by his dog, represented by the constellation **Canis Major**, and who is fighting a bull, represented by **Taurus**.



International Astronomical Union (IAU)

Because different cultures used different names and boundaries for constellations, the **International Astronomical Union (IAU)** set official boundaries to keep the sky organised.

The IAU divides the entire sky into **88 official constellations**, each covering a specific region of the sky.

More about Constellations

Two well-known patterns are the **Big Dipper** and the **Little Dipper**. The Big Dipper is part of the constellation **Ursa Major**, and the Little Dipper is part of **Ursa Minor**. A very important star called the **Pole Star (Polaris)** lies near the end of the Little Dipper's handle and appears almost fixed in the northern sky, making it useful for finding north in the Northern Hemisphere.



The Pole Star appears nearly stationary because it is located almost exactly above Earth's North Pole.

Cultural names and stories vary: in India the Big Dipper is called **Saptarīṣhi** and the Pole Star is known as **Dhruva tāṛā**. Different communities interpret the same star pattern in many ways—for example, some Central Indian tribes see the four main stars of the Big Dipper as a "**grandmother's cot**", while fishermen on the Konkan coast imagine a boat.

Night Sky Watching

To see stars and constellations well, choose nights that are clear and moonless if possible. Urban areas often hide stars because of artificial lighting, smoke and dust in the air. Open, dark locations with little light pollution give the best views of the night sky.



Light pollution from streetlights, buildings and other sources reduces the number of visible stars. Rural or high-altitude locations usually offer clearer views.

Buildings, trees and hills can block parts of the sky, so look for an open view of the horizon when possible.

Some constellations and stars cannot be seen from every place on Earth or at all times of year; which stars are visible depends on your latitude and the season. For instance, the **Pole Star** cannot be seen from the southern part of Earth. To identify a star or a constellation, you need to know what it looks like and where to find it in the sky.

Sky maps, printed star charts and mobile apps (that show the sky for your location and time) make it easier to identify stars and constellations.

Understanding Light Pollution

Light pollution is increasing rapidly and makes it harder to observe and study the night sky.

Some areas have designated **dark sky reserves and parks** where lighting is controlled to preserve dark nights for research and stargazing.

Several **organisations** work to teach people how to reduce unnecessary lighting and protect the night sky.

Apps for Identifying Stars, Constellations, and Planets

Sky Map - a mobile app that helps identify stars, constellations and planets by pointing your phone at the sky.

Stellarium - an app and free desktop program that shows a realistic sky map for any time and location, with many useful features for learning the night sky.

Our Solar System

The **Solar System** is a family of objects bound by gravity to the **Sun**. The Sun lies at the centre and the other objects-planets, moons, asteroids and comets-orbit around it.



The Sun

The Sun is a **star** and the closest star to Earth. It is a giant, extremely hot ball of gases that produces energy by nuclear reactions in its core.

The Sun provides almost all the light and heat needed for life on Earth and drives processes such as the climate, seasons, weather, the water cycle and winds.

Although the Sun is about 100 times wider than Earth, it appears much larger than other stars because it is much closer to us. Other stars are farther away, which is why they look like tiny points of light in the sky, even though some of them are much larger than the Sun.

During the day, the Sun's brightness makes it difficult to see these other stars.

Besides the Sun and Earth, there are many other objects in the sky that, along with Earth, form the **Solar System**. Most of these objects move around the Sun in a path called

revolution.



Some facts about the Sun

The average distance from Earth to the Sun is about **150 million kilometres**, commonly called one **astronomical unit (au)**. Scientists use the astronomical unit to describe distances within the Solar System.

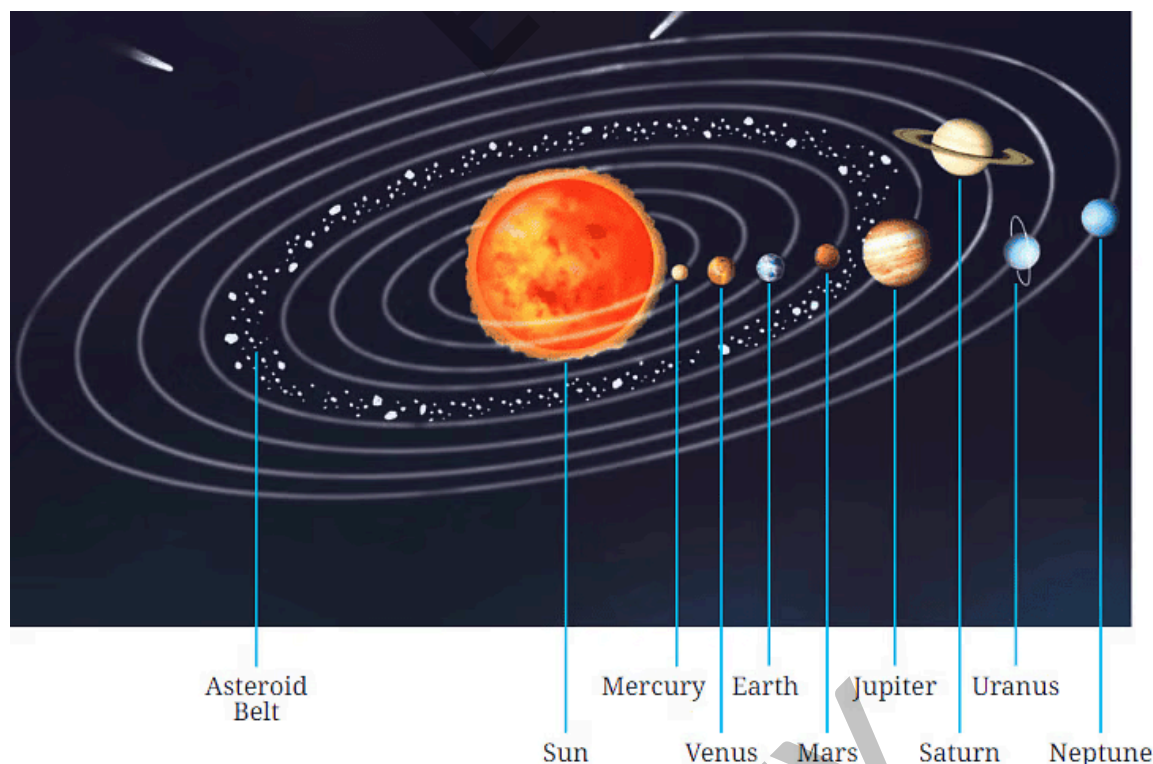
The Sun is the **brightest** object in our sky and was worshipped by many ancient cultures. Ancient people in many cultures, including India, worshipped the Sun as a **god**. In India the Sun is called **Sūrya**.

Sunlight is essential for plant growth (photosynthesis), which supports food chains and produces oxygen for living organisms.

Additionally, the Sun plays a key role in **creating climate, seasons, weather, the water cycle, and winds**. All of these factors are vital for sustaining life on Earth.

Planets

A **planet** is a large, nearly round object that orbits the Sun and does not produce its own light; it shines because it reflects sunlight. It usually follows a nearly regular path called its orbital **revolution**.



There are eight planets in our Solar System, listed in order from the Sun: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus** and **Neptune**.

The four planets closest to the Sun (**Mercury, Venus, Earth, Mars**) are smaller and have rocky surfaces.

Venus is often seen very bright in the morning or evening and is called the Morning Star or Evening Star despite being a planet.

Mars is called the Red Planet because of the reddish colour of its soil.

Earth appears blue from space because large areas of its surface are covered by water; this is why it is sometimes called the **Blue Planet**.

The four outer planets (**Jupiter, Saturn, Uranus, Neptune**) are large gas planets and have many moons; some are surrounded by rings made of dust and rock particles.

Generally, planets farther from the Sun are colder, but a planet's atmosphere can trap heat (the greenhouse effect) and change its surface temperature. For example, **Venus** is hotter than **Mercury** because its thick atmosphere traps heat.

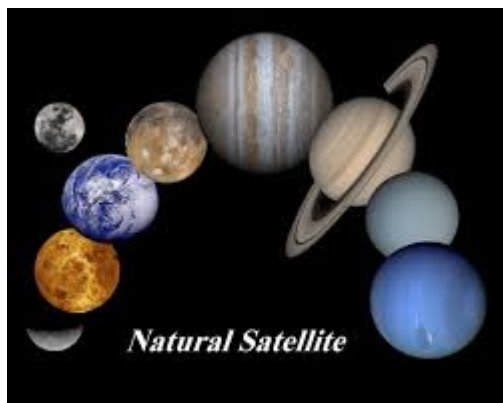
After the Sun and the Moon, **Venus** is the brightest object in our sky. Mercury, Mars, Jupiter and Saturn can also be seen without a telescope, but they look like bright points of light.

Pluto

Pluto orbits the Sun beyond Neptune. It is smaller than Earth's Moon. When Pluto was discovered it was classified as a planet, but in **2006** the **IAU** revised the definition of a planet. Under the new definition, Pluto and similar bodies are classified as **dwarf planets**.

Natural Satellites

A **natural satellite** is an object that orbits a planet. Moons are natural satellites.



Natural satellites are smaller than the planets they orbit and move around those planets because of gravity.

Earth has one natural satellite, called the **Moon**. Mars has two small moons, while the giant outer planets like **Jupiter, Saturn, Uranus, and Neptune**, have many moons of different sizes.

The Moon

The Moon is Earth's nearest celestial neighbour and takes about **27 days** to orbit Earth. The Moon has very little atmosphere, so its surface shows many round, bowl-shaped depressions called **craters** formed by meteorite impacts. Because there is no air or water to erode these craters, they remain visible for a long time.

The average distance from the Earth to the Moon is about **384,000 kilometres**.



Humans have explored the Moon using spacecraft and landers. India has launched a series of lunar missions called **Chandrayaan** to study the Moon.

Chandrayaan-1 was launched in **2008**.

Chandrayaan-2 was launched in **2019**.

Chandrayaan-3 was launched in **July 2023**. Its Vikram lander, carrying the Pragyan rover, successfully landed on the Moon on **23 August 2023**, making India the first country to achieve a soft landing near the Moon's south pole. The Government of India declared **23 August** as **National Space Day**.

A future mission, **Chandrayaan-4**, is being planned with goals that include returning soil and rock samples from the Moon.



MULTIPLE CHOICE QUESTION

Try yourself: Which planet is known as the Red Planet?

A Mercury

B Venus

C Earth

D Mars

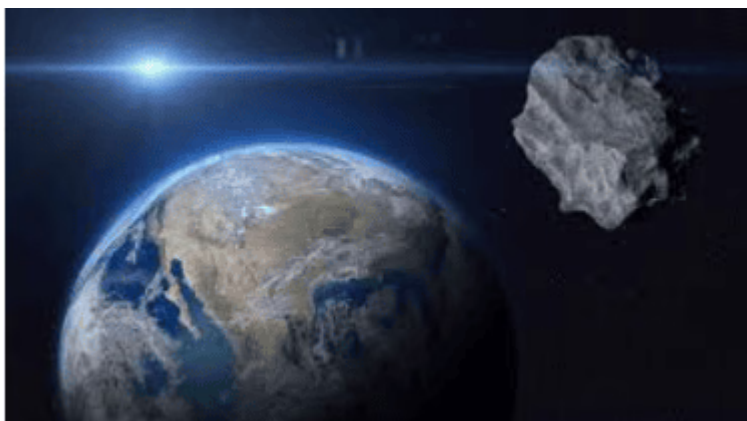
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A satellite is something that moves around a **larger object in space**. For example, the Earth is a satellite of the Sun because it orbits around the Sun.

Asteroids

Asteroids are small, rocky objects that orbit the Sun. They usually have irregular shapes rather than being spherical.

Many asteroids are found in the **asteroid belt**, a region between the orbits of Mars and Jupiter, but some follow paths that bring them near Earth.



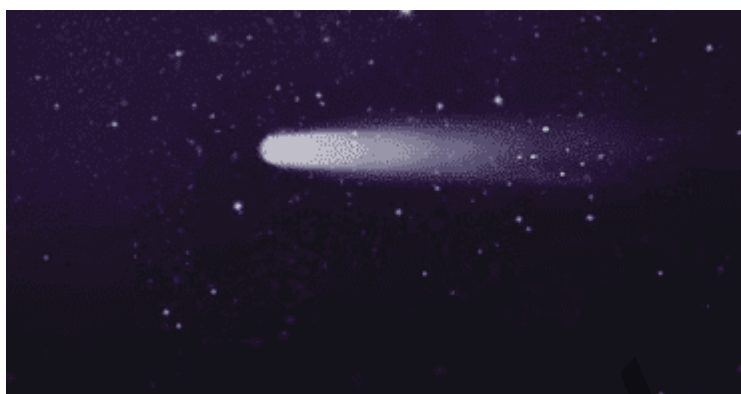
Comets

Comets are icy bodies made of dust, rock and frozen gases that travel through the outer Solar System. When they approach the Sun, heat causes the ice to vaporise and release dust, forming a glowing **coma** (a temporary atmosphere) and a tail that always points away from the Sun.

Comets far from the Sun are faint and hard to see; some comets return on regular schedules while others may travel away or break apart.

Some comets have regular orbits that bring them close to the Sun periodically, while others may escape the Solar System entirely.

Some comets can break apart, crash into the Sun, or collide with planets as they travel through space.



Halley's Comet

Halley's Comet is a famous comet that returns to the inner Solar System approximately every **76 years**.

The last time Halley's Comet was visible from Earth was in **1986**.

What Makes Up the Solar System?

The Solar System consists of the **Sun, eight planets, their moons, and many smaller bodies such as asteroids, comets and dwarf planets.**

The Sun is the largest and most massive object in the Solar System and provides the energy that makes the planets and other bodies visible by reflecting sunlight.

The Milky Way Galaxy

On a moonless night at a dark location away from city lights, you may see a faint, milky band of light stretching across the sky. This band is the **Milky Way Galaxy or Āk.āśha Gangā**, our home galaxy, which contains billions of stars. Our Solar System is only a tiny part of the Milky Way.



The Universe

Beyond the Milky Way there are countless other galaxies. Scientists study galaxies to learn how stars and galaxies form and change over time.

One of the major scientific questions is whether life exists elsewhere in the universe. To look for life, astronomers study **exoplanets**, which are planets that orbit stars other than the Sun. So far there is no confirmed evidence of life on other planets, but research and observations continue with telescopes and space missions.



MULTIPLE CHOICE QUESTION

Try yourself: Which celestial objects are made of dust, gas, rocks, and ice?

A Planets

B Comets

C Asteroids

D Moons

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Key Concepts

Asteroid: A small, rocky object orbiting the Sun, most commonly found in the asteroid belt between Mars and Jupiter.

Moon: A natural satellite that orbits a planet.

Solar System: The Sun and all the objects that orbit it, including planets, moons, asteroids and comets.

Constellation: A group of stars forming a recognisable pattern.

Comet: An icy body that releases gas and dust when heated by the Sun, forming a glowing coma and tail.

Milky Way Galaxy: The galaxy that contains our Solar System.

Planet: A large object that orbits a star and shines by reflected light.

Star: A massive, luminous sphere of hot gas that emits light and heat.

Revolution: The motion of an object around another object, for example a planet around the Sun.

Satellite: An object that orbits a planet; can be natural (a moon) or artificial (a man-made satellite).

Sun: The star at the centre of our Solar System that provides light and heat to the planets.

Summary

The night sky contains many objects-from nearby neighbours like the Moon to distant stars and galaxies. Observing stars and constellations connects cultural stories with practical navigation. The Solar System, with its Sun, planets, moons, asteroids and comets, is only a tiny part of the vast Milky Way and the much larger universe. Tools such as star charts, mobile apps and controlled dark-sky areas help us explore and understand the sky more clearly.